

e-CAM_YUV-GMSL

Release Notes



Version 1.0

e-con Systems

8/14/2024

Disclaimer

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Introduction

The e-CAM_YUV-GMSL_Products camera board is designed and developed by e-con Systems, a leading Embedded Product Design Services Company which specializes in advanced camera solutions. This camera board targets the NVIDIA® Jetson AGX Orin™ development kit.

This e-CAM_YUV-GMSL_PRODUCTS supports the following list of products STURDeCAM20_CUOAGX, STURDeCAM21_CUOAGX, NileCAM20_CUOAGX, NileCAM21_CUOAGX, NileCAM25_CUOAGX and NileCAM81_CUOAGX.

NileCAM20_CUOAGX/ STURDeCAM20_CUOAGX is a 2 MP GMSL camera and is the successor product of the NileCAM family of Serial Link cameras or serializer-deserializer (SerDes) cameras. It is a multi-board solution using AR0230 CMOS image sensor from onsemi™ along with on-board ISP for the camera module board and GMSL SerDes interface board.

NileCAM21_CUOAGX/ STURDeCAM21_CUOAGX is a 2 MP GMSL camera is the successor product of the NileCAM family of serial link cameras or serializer-deserializer (SerDes) cameras. It is a multi-board solution using AR0233 CMOS image sensor from onsemi™ along with on-board ISP for the camera module board and GMSL SerDes interface board.

NileCAM25_CUOAGX is a 2.3 MP GMSL camera and is the successor product of the NileCAM family of serial link cameras or serializer-deserializer (SerDes) cameras. It is a multi-board solution using AR0234 CMOS image sensor from onsemi™ along with on-board ISP for the camera module board and GMSL SerDes interface board.

NileCAM81_CUOAGX is an 8 MP Gigabit Multimedia Serial Link (GMSL) camera and is the successor product of the NileCAM family of Serial Link cameras or serializer-deserializer (SerDes) cameras. It is a multi-board solution using AR0821 CMOS image sensor from onsemi™ along with on-board ISP for the camera module board and GMSL SerDes interface board.

Release Details

This section describes about identifying the deliverable.

Release Version: R03_RC1

- Release Date: **13-AUG-2024**
- Product: **e-CAM_YUV_GMSL_PRODUCTS**

Description	Jetson AGX Orin™
Host machine version required for flashing software onto Jetson™ Development kit.	Ubuntu 22.04 (64-bit) or Ubuntu 20.04 (64-bit).
Sample rootfs Ubuntu operating system to run on Jetson™ Development Kit.	Ubuntu 22.04 (aarch64)
JetPack Version	JetPack 6.0
L4T version	L4T_R36.3.0
Kernel version	5.15.136-tegra
ecam_tk1_guvcview version	1.7.2-g5d361af
e-multicam version	e-multicam.elf-gdf850b2
NileCAM20 MCU Firmware Version	11NL20GMSLXXX0111043ed6b3XXXXXXX
NileCAM21 MCU Firmware Version	11NL20GMSLXXX0111064e0151XXXXXXX
NileCAM25 MCU Firmware Version	11NILE25V7XXX01110203db1cGCORAL3
NileCAM81 MCU Firmware Version	11NL81GMSLXXX01110fd8ae9bXXXXXXX

Troubleshooting

In this section, you can view the list of commonly occurring issues and their troubleshooting steps.

The systems are running in low-graphics mode error log appears on screen. How to solve this issue?

This error may occur in Jetson AGX Orin™ when complete package is flashed. To solve this issue, reset the board.

Why the camera is not detected, even if the device is connected and powered ON?

This may occur on the deserializer board when it is not powered ON. Make sure that the deserializer board is powered ON, before the Jetson™ development kit.

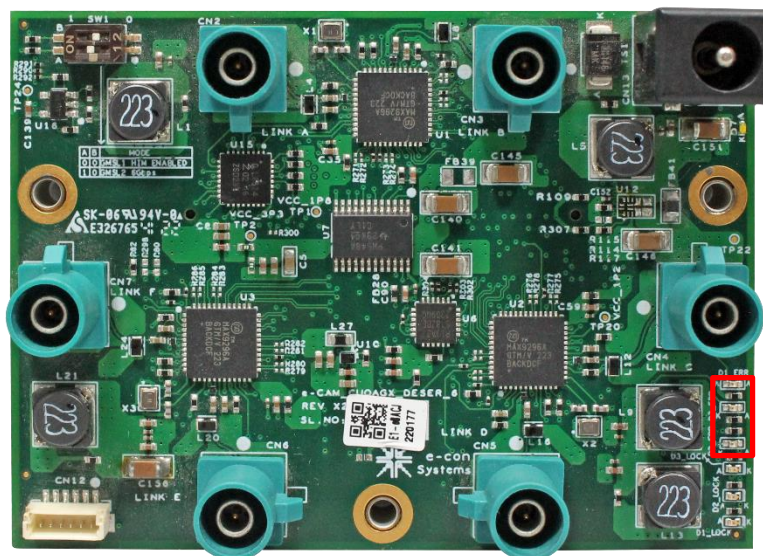
Why power ON indication LED of serializer is OFF, even if the device is connected?

When the power indicator LED of Serializer (D2) is OFF (with status LED of base board is ON), it means that power is not applied from base board. Check if the coaxial cable is inserted properly on either end.

Why error LED of deserializer is ON, if device is connected?

Apart from Lock LED, the deserializer board has a RED error LED (D1). The LED glows, when the error in the GMSL data exceeds the set threshold. The glowing of RED LED does not affect the camera streaming.

The following figure shows the Error LED (RED) on Deserializer board.



FAQ

1. **Does e-CAM_YUV-GMSL_PRODUCTS camera work with any coaxial GMSL cable available in the market?**

Currently, e-con Systems provide a very flexible and robust 3m/15m coaxial GMSL cable along with this kit. For more lengths and customizations, please write to camerasolutions@e-consystems.com with your requirement.

2. **Does e-CAM_YUV-GMSL_PRODUCTS camera support NVIDIA® TX2i?**

Currently, this camera supports NVIDIA® Orin™ AGX only. For customization, please write to camerasolutions@e-consystems.com with your requirement.

3. **Does e-CAM_YUV-GMSL_PRODUCTS camera support OpenCV?**

e-CAM_YUV-GMSL_PRODUCTS works using V4L2 APIs and is V4L2 compliant. So, any V4L2 based application can be used to access this camera. OpenCV is also compatible since it uses V4L2 to access the camera.

Please refer to https://www.e-consystems.com/Articles/Camera/accessing_cameras_in_opencv_with_high_performance.asp for detailed information about OpenCV support in e-con Systems cameras.

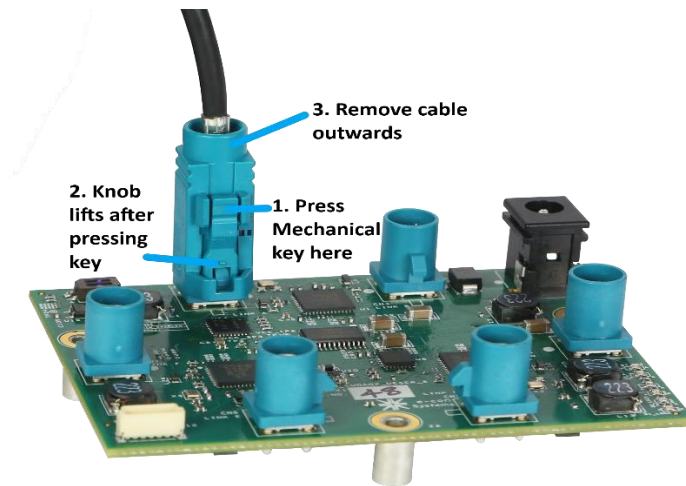
4. **How can I get the updated package?**

Please login to the [Developer Resources](#) website and download the latest release package.

5. **How to remove the coaxial cable from FAKRA plug?**

Care must be taken while removing the coaxial cable from the PCB. The FAKRA jack on coaxial cable locks with mechanical key of the FAKRA plug on board. The lock between plug and jack needs to be released, to remove the cable. On pressing the jack at the specified location below, the knob lifts and the cable can be removed.

The following figure shows the stages of removing FAKRA jack (coaxial cable) from PCB.



6. How can I mount the e-CAM_YUV-GMSL_PRODUCTS according to my application?

e-CAM_YUV-GMSL_PRODUCTS can be mounted in various ways depending on the application. Refer the link below to follow the mounting procedure.

<https://www.youtube.com/watch?v=n26FG1VbR04&feature=youtu.be>

What's Next?

After understanding the Release Notes, you can refer to the following documents to understand more about e-CAM_YUV-GMSL products.

- *E-CAM_YUV-GMSL Getting Started and Quick developer guide*
- *Respective products's Linux App User Manual*

Contact Us

If you need any support on e-CAM_YUV-GMSL products, please contact us using the Live Chat option available on our website - <https://www.e-consystems.com/>

Creating a Ticket

If you need to create a ticket for any type of issue, please visit the ticketing page on our website - <https://www.e-consystems.com/create-ticket.asp>

RMA

To know about our Return Material Authorization (RMA) policy, please visit the RMA Policy page on our website - <https://www.e-consystems.com/RMA-Policy.asp>

General Product Warranty Terms

To know about our General Product Warranty Terms, please visit the General Warranty Terms page on our website - <https://www.e-consystems.com/warranty.asp>

Revision History

Rev	Date	Description	Author
1.0	14-Aug-2024	Initial Draft	Camera Dev Team